

They have also been invited by Sharjah Municipality for a unique use case for Bandicoot that involves inspection of manholes and sample collection across the city with their robot. They are also set to participate in the upcoming Dubai Accelerator Program.

Li-Fi Solutions By Velmenni Labs

The number of connected devices across the globe is growing steadily, and is projected to cross 20 billion by 2020, according to a report by Gartner. Most devices use either Wi-Fi across 2.4GHz and 5.0GHz bands or cellular networks through respective wave frequencies utilised by network providers. Licensed bands are slowly becoming saturated, and will soon be filled to the brink, given the massive number of smart devices being used year-on-year. To find alternatives to this existing challenge, Deepak Solanki founded Velmenni R&D Labs in New Delhi to study the power of light in connectivity.

Speaking on saturating network bands, Solanki says, "Currently radio frequency (RF)-based wireless infrastructure is facing spectrum crunch due to the adoption of the Internet of Things (IoT)/connected devices. To cater to the demand generated by these devices, the government needs to license new spectrum. This is a long and tedious process globally."

The visible light spectrum (430THz to 770THz frequency) has a much larger bandwidth compared to Wi-Fi and cellular spectra. This spectrum is essentially untapped and has a lot of potential to accommodate the rising number of smart devices. This new-age connectivity technology is called Light Fidelity (Li-Fi), where data is coded inside light packets and transmitted through a Li-Fi-enabled luminary. A compatible receiver receives the light and through associated



Team Velmenni

components decodes and represents the data.

The Velmenni team is leveraging this property of light through its solutions. Solanki says, "Our solutions provide connectivity using free and unlicensed spectrum that light provides. This avoids complications surrounding licensing and various other associated processes."

Li-Fi solution and deployment

Velmenni team has developed Li-Fi access points and Li-Fi receivers that can be utilised to create a wireless network in indoor as well as outdoor scenarios using Li-Fi-enabled luminaries. The solutions are being tested through live pilot deployments and proofs-of-concept (PoCs).

The team deployed its pilot in Tallinn, Estonia back in 2014. The project was aimed at creating a wireless network infrastructure for IoT-based applications in an industrial environment. Solanki explains the setup, "We utilised the LED lighting infrastructure from the industrial setup, and created a retrofit Li-Fi solution. We integrated our Li-Fi access points with the light source and Li-Fi receivers with different tools and machines, which the factory could use. Data generated by smart tools was pushed to the cloud using Li-Fi network."

The complete setup provided reliable and uninterrupted connectivity among devices. Lights fitted at regular intervals across the workspace ensured consistent network reach across all corners.

Li-Fi also adds an extra layer of security, as data coupled in light packets is extremely difficult to access

without authorised receivers. This greatly improves data security and network access regulation among the workforce of the industry.

Solanki and team are now focusing on India with their technology, working on an array of application areas like aviation, telecom and smart buildings. They are also running several PoC projects. Solanki says, "In aviation, a use case for our solutions is wireless in-flight entertainment systems."

"In telecom, we are working with a number of service providers for improving last-mile connectivity."

"Our solution can also be used in smart lighting and connectivity within commercial spaces."

"The biggest benefit for all the above-mentioned industries is high speed, highly secure wireless connectivity."

The major challenge that the startup faced in India was to convince investors to get behind an R&D-focused idea. Solanki explains, "R&D takes time. Market adoption is slow for new technologies in India. It was challenging to find interested investors here. So, we started focusing on the international market instead and got necessary validations. Now, we are working in the Indian market as well."

Going forward, their main aim is to work towards standardisation of the technology, so it can be adopted on a mass scale. Solanki believes that down the line, once a proper standard is put in place, Li-Fi will move into the consumer space as well, where the technology can connect smartphones, tablets and other everyday devices, too. **EFY**